

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T23 · Algebraic Proof

Questions

Student book · questions with writing space

■ 10 pages

FAMILY MAP

Algebraic Proof

F01	Construct an algebraic identity proof	3 Q 8 marks
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F02	Prove divisibility or a remainder	2 Q 6 marks
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F03	Prove a universal parity statement	4 Q 13 marks
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Question 19

4MA1-2H Q14

MODULAR UNIT 2

3 marks

- 14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(3)

4MA1-2H Q18

3 marks

17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

19 $ABCED$ is a five-sided shape.

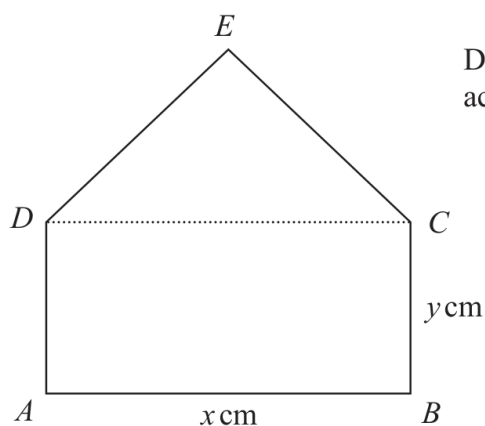


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(a) work out the value of k

$$k = \dots\dots\dots (1)$$

where a and b are integers.
Show each stage of your working clearly.

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

Show your working clearly.

(Total for Question 17 is 3 marks)

(Total for Question 16 is 2 marks)